

Utkarsh Doshi

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RESEARCH INTERESTS

Self-Supervised Learning • Representation Learning • Computer Vision • Medical Image Analysis • Generative Models • Diffusion Models • Multimodal Learning

EDUCATION

Indian Institute of Technology Hyderabad (IIT-H)

B.Tech, Electrical Engineering | CGPA: 8.17/10

Nov 2020 – Jul 2024 | Hyderabad, India

Relevant Coursework:

Machine Learning for Signal Processing • Deep Learning • Image & Video Processing • Convex Optimization • Probability & Random Processes • Information Theory • Digital Signal Processing • Communication Systems

RESEARCH EXPERIENCE

CARLAB, University of Pittsburgh

Machine Learning Researcher | Aug 2024 – Present | Pittsburgh, USA

- Research on **deep learning for medical imaging**, focusing on disease classification, segmentation, and quantitative biomarker extraction
- Collaborating with clinicians to develop AI systems for **OCT-based retinal analysis and volumetric medical imaging**
- Investigating robustness and generalization in **data-limited medical vision settings**

- Developing CNN and transformer-based architectures for dense prediction tasks
 - Stack: PyTorch, Python, Azure, Scikit-learn
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IIT Hyderabad – Research Project (with University of Pittsburgh collaboration)

Unsupervised Retinal Layer Segmentation (OCT B-scans) | Feb 2023 – May 2023

- Developed a **multi-stage unsupervised segmentation pipeline** for retinal layer extraction in OCT images
 - Integrated ESRGAN-based super-resolution, DexiNed edge detection, and denoising autoencoders
 - Collaborated with ophthalmology researchers at University of Pittsburgh for medical validation
 - Focused on improving edge quality and robustness in noisy imaging conditions
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PUBLICATIONS

First Author (Under Review)

- Generalizable Multimodal Image Registration via Label-free Vessel Segmentation

Co-Author (Under Review)

- Benchmarking Diffusion Models for OCT Fluid Biomarker Segmentation

Under Review

First Author

- Cross-Device Transfer Learning for Volumetric OCT Classification of AMD
- Comparative Evaluation of Deep Learning Models for GA Segmentation in En-face OCT Images

Co-Author

- Deep Learning-Based Automated Segmentation and Quantification of Retinal Structures in OCT
 - Enrichment Ratio Analysis for Recurrence Prediction in Central Serous Chorioretinopathy
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INDUSTRY EXPERIENCE

Unstudio.ai

Machine Learning Intern | Dec 2023 – Jan 2024 | Remote

- Optimized inference pipelines for **Stable Diffusion XL** models
 - Implemented and trained **LoRA and T2I adapters** for controllable image generation
 - Improved model efficiency for diffusion-based workflows
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Philips Innovation Campus

Machine Learning Engineer Intern | May 2023 – Jul 2023 | Bengaluru, India

- Developed CNN-based system for **lung nodule classification (5-class problem)**
 - Achieved 96% accuracy on imbalanced medical dataset
 - Evaluated ResNet and VGG architectures for clinical imaging pipeline
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FrinksAI

Machine Learning Intern | Dec 2022 – Mar 2023 | Remote

- Built and evaluated YOLOv5 and YOLOv8-based pipelines for segmentation tasks
 - Designed annotation pipeline converting binary masks into YOLO format
 - Developed dataset pipelines using Roboflow for production training workflows
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Snazzy (YC-backed)

AI Intern | Jun 2020 – Jul 2020 | Remote

- Implemented facial transformation model based on research paper for smile prediction
 - Built dataset of 600+ images using Selenium-based web scraping
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PROJECTS

Machine Unlearning (NeurIPS-style project)

IIT Hyderabad | Oct 2023 – Nov 2023

- Ranked top 5% in course-based Kaggle competition
 - Developed efficient **machine unlearning algorithm** for removing influence of forget-set samples
 - Explored parameter-efficient fine-tuning and low-rank adaptation methods
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VAE-based Image Generation (Personal Research Project)

- Implemented Variational Autoencoders for representation learning and image synthesis
 - Explored latent space interpolation and attribute vector arithmetic using CelebA dataset
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Medical Image Segmentation (U-Net / E-Net)

- Implemented segmentation models from research papers for medical and drone imagery
 - Compared lightweight E-Net vs U-Net tradeoffs for efficiency vs accuracy
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CAPTCHA Decoder System

- Developed CNN-based CAPTCHA decoding system for university system automation
 - Achieved 800+ downloads in internal deployment
 - Built full pipeline from dataset creation to inference
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TECHNICAL SKILLS

Machine Learning & AI:

Deep Learning, Computer Vision, Self-Supervised Learning, Representation Learning, Medical Image Analysis, Segmentation, Diffusion Models

Frameworks & Tools:

PyTorch, Python, Scikit-learn, OpenCV, Azure, TensorFlow (basic), Docker (basic)

ACHIEVEMENTS

- 3rd place, IEEE Video & Image Processing Cup (Georgia Tech collaboration), \$1500 prize
 - Ranked AIR 1586 in JEE Advanced (top ~1%)
 - 99.4 percentile in JEE Mains
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LANGUAGES

English (Professional) • Hindi (Native)